

RLHF Can Speak Many Languages: Unlocking Multilingual Preference Optimization for LLMs

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Abstract

Preference optimization techniques have become a standard final stage for training state-of-art large language models (LLMs). However, despite widespread adoption, the vast majority of work to-date has focused on first-class citizen languages like English and Chinese. This captures a small fraction of the languages in the world, but also makes it unclear which aspects of current state-of-the-art research transfer to a multilingual setting.

In this work, we perform an exhaustive study to achieve a new state of the art in aligning multilingual LLMs. We introduce a novel, scalable method for generating high-quality multilingual feedback data to balance data coverage. We establish the benefits of cross-lingual transfer and increased dataset size in preference training. Our preference-trained model achieves a 54.4% win-rate against Aya 23 8B, the current state-of-the-art multilingual LLM in its parameter class, and a 69.5% win-rate or higher against widely used models like Gemma, Mistral and Llama 3. As a result of our efforts, we expand the frontier of alignment techniques to 23 languages, covering approximately half of the world’s population.

1 Introduction

What languages are favored in technological progress is often deeply intertwined with historical patterns of technology access and resources (V et al., 2020; Bird, 2022; Singh et al., 2024). Preference optimization is a valuable and widely adopted post-training technique to align large language models (LLMs) with human preferences (Christiano et al., 2017b; Stiennon et al., 2022; Ouyang et al., 2022a; Bai et al., 2022). It has also been shown to lead to large gains in performance across a wide variety of NLP tasks (Wang et al., 2024; Ivison et al., 2023; Xu et al., 2024; Lightman et al., 2024). To-date, the majority of progress in preference optimization has over-fit to a small handful of

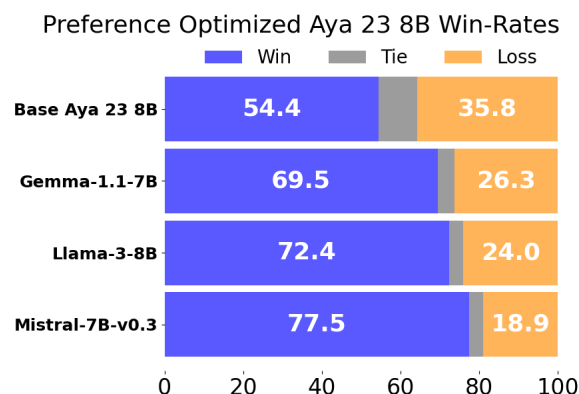


Figure 1: **Win-rates between our preference-trained model with the other state-of-the-art open weight models:** Our preference-trained model based on Aya-23-8B significantly outperforms the original Aya-23-8B, Gemma-1.1-7B-it, Meta-Llama3-8B-Instruct, and Mistral-7B-Instruct-v0.3.

languages, resulting in large gaps in performance outside of English (Schwartz et al., 2022; Kotek et al., 2023; Khandelwal et al., 2023; Vashishtha et al., 2023; Khondaker et al., 2023), and also risks introducing security flaws for all users (Yong et al., 2024; Nasr et al., 2023; Li et al., 2023a; Lukas et al., 2023; Deng et al., 2023).

While expanding the frontier of what languages are supported by AI is an increasingly urgent problem, extending preference optimization to a multilingual setting is a non-trivial challenge. First, numerous works have shown that multilingual modeling typically faces both a data scarcity and data quality problem (Singh et al., 2024; Üstün et al., 2024; Dodge et al., 2021; Kreutzer et al., 2022; Lucioni and Viviano, 2021). This is even more pronounced for high-quality preference data which is virtually non-existent in many languages. Collecting multilingual preference datasets through human annotation is expensive and time intensive (Boubdir et al., 2023; Chaudhari et al., 2024), and while prior works have proposed the use of LLMs to

synthetically create preference datasets (Bai et al., 2022; Yuan et al., 2024; Pace et al., 2024), these efforts predominantly focus on English. The few efforts that have focused on multilinguality have relied on translation, introducing artifacts and resulting in a lack of diversity in preference pairs (Lai et al., 2023), which is known to be critical to model performance (Naik et al., 2023; Chung et al., 2023; Li et al., 2023b; Lahoti et al., 2023; Kirk et al., 2024a).

Second, preference optimization in many languages simultaneously is a difficult machine learning task. The lack of studies in preference optimization outside of English raises questions on how findings from monolingual optimization would transfer. Training dynamics of RLHF are known to be often unstable (Casper et al., 2023; Gao et al., 2022a; Chaudhari et al., 2024), which can be exacerbated by the involvement of multiple languages where preference data is from a heterogeneous distribution and negative transfer between languages is possible (Wang et al., 2020, 2019). The few existing works on multilingual RLHF (Lai et al., 2023, 2024) exhibit poor results and are outperformed by massively multilingual language models without any preference optimization (Üstün et al., 2024; Aryabumi et al., 2024).

Poor performance of the few existing multilingual preference optimization works begs the question: *Is this a result of fundamental limitations with standard preference optimization techniques (especially in heterogeneous optimization settings like multilingual) or whether we are lacking high quality multilingual data?*

In this work, we exhaustively study the aforementioned challenges. Our goal is to systematically understand key variables which might impact multilingual alignment, including the source and amount of available preference data, offline vs online RLHF techniques, and the effect varying number of languages covered in preference optimization training data. We complete a comprehensive set of experiments with state-of-the-art alignment techniques including DPO (Rafailov et al., 2023) and RLOO (Kool et al., 2019; Ahmadian et al., 2024) starting from the 8-billion-parameter instruction-finetuned Aya model covering 23 languages (Aya-23-8B; Aryabumi et al., 2024). Our findings can be summarized as follows:

1. **Preference optimization exhibits cross-lingual transfer.** We show that preference-

optimization even with only English data improves performance in other languages. However, the addition of a few more languages significantly increases cross-lingual transfer, achieving win-rates on unseen languages up to 54.9% when including 5 languages in training data compared to 46.3% when training only on English.

2. **Multilingual preference data is necessary for aligning multilingual LLMs.** We find that increasing the number of languages in preference optimization training data consistently improves multilingual performance compared to English-only training data, increasing win-rates by up to 7.0% from 46.4% to 53.4% when all languages are included.
3. **Online preference optimization outperforms offline optimization.** RLOO as an online method achieves better overall performance than DPO by a maximum 10.6% difference in their average win-rates (54.4% vs 43.8%). Furthermore, we find that RLOO also enables better cross-lingual transfer, achieving up to 8.3% increase over DPO in average win-rate on languages not included in training.
4. **Preference optimized Aya 23 8B outperforms other open weights models** Preference optimization leads to large gains in win-rates against both the original Aya model (54.4% win-rate) and widely used models including Meta-Llama3-8B-Instruct (AI@Meta, 2024) (72.4% win-rate), Gemma-1.1-7B-it (Gemma-Team, 2024) (69.5% win-rate), Mistral-7b-Instruct-v0.3 (77.5% win-rate) (Jiang et al., 2023) across all 23 languages as shown in Figure 1.¹

2 Methodology

2.1 Addressing Data Scarcity

The limited prior work on multilingual preference training involved fully translated preferences from English (Lai et al., 2023), however the Okapi model has since been outperformed by non-preference aligned models including the base Aya 23 8B model we experiment with here (Aryabumi et al., 2024). We hypothesize that the poor performance may be

¹We will release our preference-trained model upon publication.

due to the reliance on translated preferences. While language coverage may be improved by translation (Ranaldi and Pucci, 2023; Üstün et al., 2024), the introduction of translation artefacts known as *translationese* (Bizzoni et al., 2020; Vanmassenhove et al., 2021) can hurt performance. Furthermore, repeatedly translating the same preference pairs can hurt preference diversity. The extract trade-off between the positive and negative benefits is not well understood and is difficult to isolate empirically (Yu et al., 2022; Dutta Chowdhury et al., 2022)

Here, we attempt to avoid some of the issues with translation by creating preference pairs that intentionally aim to steer model generations away from *translationese*. First, we construct a diverse set of general instruction-following multilingual prompts by translating approximately 50K English prompts from ShareGPT² into the remaining 22 languages supported by Aya 23 8B. Automatic translation is done by using NLLB 3.3B (NLLB Team et al., 2022).

Source of completions After translating prompts, we generate completions for each language by using multiple LLMs of varying multilingual capability. This ensures increased completion diversity versus the alternative of simply translating the original English preference models. More specifically, we use Cohere’s Command³ and Command R+⁴ models, where the latter is explicitly trained for multilingual performance.⁵ For Command, we generate English completions as the model is primarily proficient in English, and translate them into the other 22 languages. For Command R+, we generate a completion from the same prompt in-language. This method enables obtaining a pair of multilingual competitions for each prompt with varying quality based on the difference in models’ capabilities and the use of machine translation.

The use of *translated* completions and comparing with high-quality *direct* multilingual generations allows the model to steer away from translation artifacts. Note that the translated competitions are ranked as “bad completions” by 91% hence, in most cases the preference ranking acts as a proxy label for translated completions.

²<https://sharegpt.com>

³<https://docs.cohere.com/docs/command-beta>

⁴<https://docs.cohere.com/docs/command-r-plus>

⁵<https://docs.cohere.com/docs/command-r-plus#unique-command-r-model-capabilities>

2.2 Offline vs Online Preference Training

Reinforcement Learning from Human Feedback (RLHF; Christiano et al., 2017b; Stiennon et al., 2022; Ouyang et al., 2022a; Bai et al., 2022) proposed as the first framework for aligning language models to human preferences, has become a key for training state-of-the-art LLMs (OpenAI et al., 2023; Touvron et al., 2023; Anthropic, 2024; Reid et al., 2024). Canonically, PPO (Schulman et al., 2017b) has been used in RLHF as the online RL algorithm (Stiennon et al., 2022; Ouyang et al., 2022b; Nakano et al., 2021). However, recent offline methods such as Direct Preference Optimization (DPO) (Rafailov et al., 2023) and subsequent works in the same direction (Azar et al., 2024; Ethayarajh et al., 2024; Choi et al., 2024), have proven increasingly popular due to reduced computational complexity. Traditional online methods such as PPO and REINFORCE (Williams, 1992) require an additional network in addition to the policy, maintaining a reward model in memory, and also using the policy to generate doing training, all of which DPO does not require as it is fully offline.

A fractured experimental ecosystem and non-standardized datasets have made it difficult to evaluate the relative merits of both approaches comprehensively. However, recent work in an English setting (Tajwar et al., 2024; Tang et al., 2024) suggests that although the DPO-family of methods theoretically optimize the same objective as traditional RL algorithms, they under-perform compared to well-tuned traditional online RL methods *due to the lack of online generations (on-policy or off-policy) and critique as provided by the reward model*. (Tajwar et al., 2024) also provides empirical evidence that explicit negative gradients during training improve over online methods without them. Given that multilingual datasets are far more heterogeneous than most datasets, it is unclear how these findings apply to massively multilingual settings.

Additionally, it is of great interest what method most benefits from cross-lingual transfer to unseen languages. Thus, in addition to **DPO**, we benchmark **REINFORCE-Leave-one-out (RLOO)** (Kool et al., 2019; Ahmadian et al., 2024). Ahmadian et al. (2024) shows that PPO may not be the right tool for RLHF, and that simpler REINFORCE-style methods such as Vanilla Policy Gradient and RLOO, are competitive or outperform PPO. In their experiments, RLOO outperforms both PPO and DPO, and incurs significantly

lower computational overhead compared to PPO. Additionally, it has a contrastive loss by nature, which (Tajwar et al., 2024) improves learning compared to traditional RL. In Appendix Section A, we give a brief background and introduction to each method.

Preference data mixtures To evaluate if the multilingual preference data is essential in all languages and also to measure the cross-lingual transfer during preference training, we design various preference data mixtures where we control the number of languages covered and the amount of the preference data per language:

1. English-only: English-only preference data mixture that includes 50K prompts with ranked generations. We term this this variant EN-1-50K and use it to understand **cross-lingual benefits that accrue from English-only preferences**. This is important, given it is the standard formulation of research to date.

2. 5 language subset: Multilingual mixture that includes English, Vietnamese, German, Turkish, and Portuguese with a total amount of 50K prompts (10K per language). These 5 languages contain a mixture of higher and lower resource languages, and in their language families and scripts. We refer to this variant as ML-5-50K, training on only this mixture and evaluating on the remaining 18 languages allows us to measure the impact of multilingual preference data on **cross-lingual transfer to unseen languages** in comparison to English-only preference data.

3. All languages (fixed data budget): Multilingual mixture with all 23 languages supported by Aya 23 8B. This is our fixed budget variant where the total number of prompts is kept the same at 50K (approximately 2.2K examples per language) to compare with EN-1-50K and ML-23-50K. This variant **measures performance trade-offs including all languages given the same data budget**. We refer to this variant as ML-23-50K.

4. All languages (not-fixed data budget): Our most comprehensive preference data mixture where we include all 23 languages with 10K prompts per language (230K in total). This mixture which we refer to as ML-23-230K enables us to evaluate the impact of a larger preference data budget in comparison with ML-23-50K.

Agreement between RM and GPT-4

English	87.9%
Vietnamese	88.7%
Turkish	84.4%
Portuguese	91.0%
German	85.5%
Avg 23 Languages	87.3%

Table 1: Ranking agreement rate for the Reward Model used in our experiments and GPT-4 Turbo on randomly selected Multilingual Dolly responses generated from Command and Command R+ (512 randomly selected per language). Unlike the Reward Model, GPT-4-Turbo is capable of outputting ties. GPT-4-Turbo selects tie result 3.1% of the time on dataset.

3 Experimental Set-up

Multilingual Base Model We perform all experiments with Aya 23 8B (Aryabumi et al., 2024) which is chosen because it (1) is massively multilingual, pre-trained and supervised fine-tuned for 23 languages, and (2) it achieves state-of-the-art multilingual performance in 23 languages compared to other commonly used LLMs in its class, outperforming Mistral-7B (Jiang et al., 2023), Gemma-7B (Gemma-Team, 2024), and Aya-101-13B (Üstün et al., 2024). On multilingual benchmarks and open-ended generations, Aya 23 achieves a 65% win-rate or higher in head-to-head comparisons with popular open source models (Aryabumi et al., 2024). Furthermore, Aya 23 is an open weights model that is not preference-trained, allowing us to isolate the impact of multilingual preference optimization.

Reward Model We use a closed-source multilingual reward model (RM) which is competitive with top-scoring state-of-the-art RMs on the Reward-Bench Leaderboard (Lambert et al., 2024). This reward model achieves high LLM response ranking agreement with GPT-4-Turbo on English and Non-English languages as shown in Table 1. We intentionally use a separate reward model from the model we use for llm-as-a-judge evaluation (GPT-4-Turbo⁶) given the known biases incurred by using the same model for both (Verga et al., 2024; Bansal et al., 2024).

Preference Optimization Training We train Aya 23 8B model for 2 epochs in both DPO

⁶<https://platform.openai.com/docs/models/gpt-4-turbo-and-gpt-4>

and RLOO experiments. DPO runs are trained with KL-penalty $\beta = 0.5$, learning rate $5e-7$, and AdamW optimizer (Kingma and Ba, 2014). RLOO runs are trained with RLOO $k = 2$, KL-penalty $\beta = 0.01$, learning rate $5e-6$, AdamW optimizer, and online generation sampling temperature of 0.75. All runs are performed on a single node with either 8 x Nvidia A100 80GB or 8 x Nvidia H100 80GB GPUs with DeepSpeed ZeRO Stage 3 (Rajbhandari et al., 2020) and full fine-tuning of all 8 billion model parameters. We performed hyperparameter sweeps for learning rate $lr \in \{5e-8, 5e-7, 5e-6, 5e-5\}$ and for KL-penalty $\beta \in \{0.05, 0.1, 0.5\}$ for both DPO and RLOO to the best of our ability. For a fair comparison with DPO, we utilize the same reward model for RLOO which is used to generate our synthetic multilingual preference dataset (for ranking the generations).

3.1 Model Comparisons

We evaluate against multiple state-of-the-art open-source models to ensure a comprehensive evaluation. Details of each model are below:

- **Meta-Llama-3-8B-Instruct** (AI@Meta, 2024) is an 8B parameter open-source instruction fine-tuned model which has been pre-trained on over 15T tokens. Over 5% of the pretraining data is high-quality non-English data, covering over 30 languages. The model is supervised fine-tuned and preference optimized with both offline (DPO) and online (PPO) algorithms
- **Mistral-7B-Instruct-v0.3** (Jiang et al., 2023) is an open-source instruct fine-tuned edition of the Mistral-7B pre-trained model. The model is trained on instruction datasets publicly available on the HuggingFace repository.
- **Gemma-1.1-7B-it** (Gemma-Team, 2024) is a 7B parameter instruction fine-tuned model trained with Gemini models’ architectures, data, and training recipes (Gemma-Team et al., 2024) on 6T tokens of data from web documents, mathematics, and code that are primarily English. In addition to the supervised fine-tuning, this model is also further fine-tuned using RLHF on collected pairs of preferences from human annotators.

We note that while the models we evaluate do not explicitly claim to support multiple languages,

in practice, they are heavily used by multilingual users relative to explicitly multilingual models like mT0 and BLOOMZ (Muennighoff et al., 2023a). We also observe that they perform well in practice.

3.2 Evaluation

We assess the preference-optimized models on the multilingual open-ended generation and summarization tasks using LLM-simulated evaluation:

1. **Open-ended generations** For open-ended generations, we use dolly-machine-translated test set from the **Aya evaluation suite** (Singh et al., 2024) which is a held-out test set of 200 instances from the Dolly-15k dataset (Conover et al., 2023) translated into 101 languages. This test set was curated by avoiding instructions that include culturally-specific or geographic references. For languages that are available (Arabic, French, Hindi, Russian, and Spanish), we use **dolly-human-edited** test set (Singh et al., 2024), an improved version of **dolly-machine-translated** post-edited by professional human annotators to correct any translation issues.
2. **Summarization Task** For summarization, we use **XLSum** (Hasan et al., 2021), for the subset of 15 languages covered by the benchmark within the Aya 23 language coverage.

Across both tasks, we measure LLM-simulated win-rates which have been shown to be highly correlated with human evaluation for both monolingual English settings (Dubois et al., 2024) and multilingual settings (Üstün et al., 2024). We use GPT-4-Turbo as an LLM-judge and follow the same procedure described by Üstün et al. (2024). To minimize bias, we randomize the order of model outputs. The judge prompt can be found in Appendix D. For evaluation, we use max prompt context length of 512, maximum generation length of 512, and sampling temperature of 0.75.

4 Results and Discussion

Win-rates Against Open-Weights Models Figure 1 and Table 2 show the win-rates of our preference-trained models against state-of-the-art open-source models. Importantly, the base Aya 23 8B already achieves high win-rates against Gemma-1.1 (62.1%), Llama-3 (66.6%), and Mistral-v0.3 (69%) averaged across all 23 languages on Dolly open-ended generations. Preference-optimized Aya

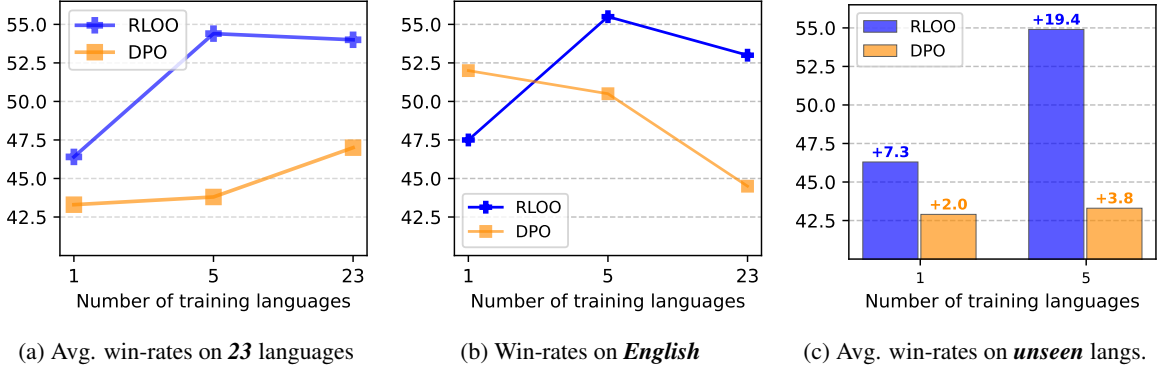


Figure 2: DPO and RLOO results with an increasing number of languages in preference training data. We report win-rates for (a) the average of 23 languages, (b) only English, and (c) the average of unseen languages, reflecting the cross-lingual transfer.

models extend this lead further. Concretely, our best variant of RLOO leads to 69.5% (+7.4), 72.4% (+5.8), and 77.5 (+8.5) win-rates against Gemma, Llama-3, and Mistral respectively.

Win Rates Against Aya-23-8B Table 3 shows win-rate results for open-ended generations and summarization respectively. Win-rates are measured against Aya 23 8B. We report win-rates for both DPO and RLOO trained with different preference data mixtures. Due to the space constraints, summarization results are shown in Appendix (Table 7). Our best variant across experiments achieves 70.7% over the base Aya model.

Online optimization vs offline We find that RLOO (online RL) outperforms DPO (offline) across the board in multilingual performance. As shown in in Table 3, RLOO achieves higher win-rates with all the preference data mixtures compared to DPO. For the 23-language mixture, RLOO achieves 54.0% win-rates whereas DPO reaches 47.0%.

Furthermore, RLOO achieves higher cross-lingual transfer with English-only and 5-language preference training in comparison to DPO. On languages unseen during training, RLOO achieves 3.4% higher win-rate compared to DPO (46.3% vs 42.9%) when training only on English and an 11.6% higher win-rate (54.9% vs 43.3) when training on the 5-language subset as shown in Table 5. This is inline with recent works (Tajwar et al., 2024; Tang et al., 2024) which suggest that online sampling outperforms fully offline methods. Additionally, the significant improvement in cross-lingual transfer with RLOO compared to DPO, suggests that online sampling also enables *generalization* be-

yond the distribution of the training prompts. This is complementary to findings of (Kirk et al., 2024b) which show that online RLHF enables better generalization than SFT.

Increasing multilinguality in preference data improves winrates

For a fixed dataset size of 50K pairwise preferences, we find that increasing the number of languages in training data improves the overall performance as shown in Figure 2a and Table 3. For DPO, win-rates against the base model on 23 languages increases from 43.3% to 47.0%.⁷ For RLOO, this gain is most visible when the number of training languages is 5 where win-rate improves from 46.4% to 54.0% (+7.6 increase). Interestingly, in open-ended generations, using all 23 languages does not improve performance further for RLOO, however, in summarization, the 23-language training mixture increases win-rates compared to the 5-language subset (from 65.1% to 70.7%, Table 7).

English also benefits from multilingual training

Our experiments also show that English can benefit from multilingual preference training and positive transfer, even when there are fewer total English examples in the training data. As shown in Figure 2b (and Table 4), despite only having approximately 2K English examples, RLOO improves English win-rates from 47.5% to 53.0% with 23-languages training data. However, English win-rates drop for DPO as the number of languages increases when there is a fixed budget of 50K examples, showing that DPO is more prone to negative interference.

⁷Win-rates that are under 50% do not correspond lower performance since our evaluation allows for *Tie*. To indicate if there is a performance gain, we also report the difference between win- and loss-rates ($\Delta W-L$) in the results.

		Average 23 Languages		
		Win%	Loss%	$\Delta W-L$ %
BASE	GEMMA-1.1	62.1	29.4	32.7
	LLAMA-3	66.6	29.4	37.2
	MISTRAL-v0.3	69.0	26.8	42.2
DPO	GEMMA-1.1	67.7	27.1	40.6
	LLAMA-3	71.0	24.7	46.3
	MISTRAL-v0.3	74.7	21.8	52.9
RLOO	GEMMA-1.1	69.5	26.3	43.2
	LLAMA-3	72.4	24.0	48.4
	MISTRAL-v0.3	77.5	18.9	58.6

Table 2: Open-ended generation (Dolly) win-rates for the base Aya 23 8B, and DPO/RLOO preference optimized Aya 23 8B models against Gemma-1.1-7B-it, Meta-llama-3-8B-Instruct and Mistral-7B-Instruct-v0.3. We report average win-rates on 23 languages for the best ML-23-230K checkpoints for DPO and RLOO.

Cross-lingual transfer to unseen languages

Preference training only with English, achieves performance gains for languages not seen in the training data as shown in Figure 2c (and Table 5). This gain ($\Delta W-L$) is 2.0% for DPO and 7.3% for RLOO. Furthermore, using a 5-language subset (ML-5) significantly increases these gains to 3.8% and 19.4% for DPO and RLOO respectively. These results provide strong evidence of cross-lingual transfer in preference optimization, which is significantly more present after online training, with an increased degree of transfer facilitated by multilingual training data.

Role of data size and reward over-optimization

For DPO, increasing the amount of multilingual data from approximately 2K to 10K per language in the 23-language mixture improves win-rates from 47.0% to 50.2% (Table 3). For RLOO, however, the same increase does not lead to an improvement. For all runs, the best checkpoint was the last one (epoch 2), except for the RLOO ML-23 230K run where we observed significant performance degradation after 0.5 epochs, which may be caused by reward model overoptimization (Gao et al., 2022b). Prior works have shown that low-resource languages can jailbreak LLMs (Yong et al., 2024) and it is likely that reward models, which usually are initialized from LLMs, likely share similar vulnerabilities. The degradation for the RLOO ML-23 230K run we observe may be caused by online optimization exploiting more languages and prompts where the reward model may be more prone to reward hacking, as this run includes all 23 languages and more prompts per language than other runs.

		Average 23 Languages		
		Win%	Loss%	$\Delta W-L$ %
DPO	EN-1	43.3	40.6	2.7
	ML-5	43.8	39.1	4.7
	ML-23	47.0	37.1	9.9
	ML-23*	50.2	39.0	11.2
RLOO	EN-1	46.4	38.9	7.5
	ML-5	54.4	35.8	18.6
	ML-23	54.0	38.0	16.0
	ML-23*	53.4	37.0	16.4

Table 3: Open-ended generation (Dolly) win-rates for DPO/RLOO preference optimized Aya models against the original Aya 23 8B. We report average win-rates on 23 languages for multiple training data mixtures: EN-1 (English Only), ML-5 (5 Languages), and ML-23 (23 Languages). All the data mixtures consist of 50K total training examples with the exception of ML-23*, which includes 230K total training examples. We report results for the best checkpoint across 2 epochs.

Is there a multilingual alignment tax? Post-training stages of LLMs including supervised fine-tuning and preference optimization have increasingly been torn between objectives: improving traditional discriminative benchmarks like HellaSwag (Zellers et al., 2019), MMLU (Hendrycks et al., 2021) and training LLMs to follow instructions, acquire conversational abilities, and be helpful and harmless (Askell et al., 2021). Recent work on multilingual instruction finetuning (Üstün et al., 2024) has found that improvements in open-ended generative tasks introduce trade-offs with discriminative tasks. However, this work only studies the tensions introduced by supervised instruction finetuning. More recent work (Tang et al., 2024) on preference training suggests that offline methods impart improved ability for discriminative tasks, whereas on-policy sampling improves generative quality. Here, we explore whether this holds for multilingual and characterize the trade-off between discriminate and generative performance.

Table C shows multilingual benchmark results of the preference optimized Aya models and the original Aya-23-8B. We follow Aryabumi et al. (2024) and evaluate models on the unseen tasks (XWinograd (Muennighoff et al., 2023b), XCOPA (Ponti et al., 2020), XStoryCloze (Lin et al., 2022)), mMMLU (Lai et al., 2023), and MGSM (Shi et al., 2023) using eval-harness framework (Gao et al., 2021).

We find that both DPO and RLOO are robust in terms of their results in multilingual benchmarks, matching the performance of the base Aya 23 8B

model. More specifically, multilingual preference optimization through both DPO and RLOO only lead to a slight drop in MGSM by an accuracy of 0.6 (36.6 vs 36.0) and 1.5 (36.6 vs 35.1) respectively. In contrast to recent work (Tang et al., 2024) in a monolingual setting, this shows that multilingual preference optimization can substantially improve generative performance as discussed in Section 4, while incurring minimal tax, given the right regularization, on common multilingual NLP tasks.

5 Related Work

Reinforcement Learning from Human Feedback (RLHF) RLHF has become the dominant paradigm for aligning LLMs to human preferences. Canonically, this involves training a reward model and using a reinforcement learning algorithm like PPO (Schulman et al., 2017a) or RLOO (Ahmadian et al., 2024) to optimize the LLM policy to maximize reward of online samples generated throughout training. (Ouyang et al., 2022b; Stiennon et al., 2020; Christiano et al., 2017a). There has been a plethora of work attempting to take the online inference aspect, and the optimization difficulties and complexities of RL that come with it, out of RLHF. The most prominent of these, is the family of methods base upon Direct Preference Optimization (DPO) (Rafailov et al., 2023), such as IPO (Azar et al., 2024), KTO (Ethayarajh et al., 2024), and SRPO (Choi et al., 2024). This family of methods directly fine-tunes an LLM to be implicitly consistent with collected preference data, forgoing the need for training a separate reward model.

Reinforcement Learning from AI Feedback (RLAIF) Collecting human feedback is often very expensive. Thus many recent works (Bai et al., 2022; Yuan et al., 2024), make use of feedback generated from AI, which can be LLMs which have been already optimized for alignment with human preferences with AI. Often these LLMs can be used to provide additional rankings, ratings, or natural language feedback, which can be used in subsequent RLHF training.

Preference Optimization for Multilingual LLMs There have been limited efforts on multilingual preference optimization to-date. (Lai et al., 2023) present a multilingual instruction tuning framework, where they preference train multilingual LLMs such as BLOOMZ (Muennighoff et al.,

2023a) for 26 non-English languages with RLHF. They synthetically generated a preference dataset by translating an extended version of the Alpaca dataset (Taori et al., 2023), generating responses from their target LLM and ranking back-translated (into English) responses with ChatGPT.⁸ In contrast to our work, they perform preference optimization for each language separately. However, due to their potentially low-quality dataset which heavily relies on translations, their resulting language-specific models are outperformed by other massively multilingual LLMs without preference optimization (Üstün et al., 2024; Aryabumi et al., 2024). Wu et al. (2024) study cross-lingual transfer in reward model (RM) training where they propose using preference data in one source language to train an RM for target language alignment. They show that RMs based on a multilingual base model exhibit zero-shot cross-lingual transfer consistently across different languages. However, they do not experiment with using multiple source languages in training, which we show is crucial in the preference optimization both for offline optimization such as DPO and online RL methods such as RLOO.

6 Conclusion

Our work presents a comprehensive study on multilingual preference optimization. We show that the inclusion of multilingual data in preference optimization leads to significant improvements in multilingual performance over English-only preference optimization. This improvement scales both with the number of languages and the total number of examples included in the training data. Additionally, we show that preference training exhibits cross-lingual transfer, leading to significant gains in languages not present in the training data. We also find that using online preference optimization outperforms offline preference optimization, highlighting the importance of online samples during training.

As a result of our study, we expand the frontier of alignment techniques to 23 languages which cover half the world’s population, by successfully preference-training an 8 Billion Aya 23 model that outperforms both the original Aya 23 8B base and widely used models including Gemma, Mistral, and Llama 3.

⁸<https://openai.com/blog/chatgpt/>

7 Limitations

A potential risk of relying on synthetic and translated datasets is the presence of particular cultural biases in model behavior. The prompts used in ShareGPT to seed the creation of the synthetic data over-index on contributions from the Global North or Western regions (Longpre et al., 2023). This could introduce a skew towards a narrow selection of cultural viewpoints.

Our preference-trained model covers 23 languages and improves performance relative to the closest open-source model. However, this is still only a tiny fraction of the world’s linguistic diversity which encompasses 7000 languages. Furthermore, in this research we don’t distinguish between dialects within the languages we cover, which are an important part of how language is used in practice (Zampieri et al., 2020; Wolfram, 1997; Brown et al., 2020; Lent et al., 2022; Blaschke et al., 2023; Falck et al., 2012). Future work, should aim to include more of the world’s population and therefore languages.

Due to compute constraints, we are limited in our ability to run preference optimization experiments for larger models. Many of the runs we describe in this work for a single run can take 5 days to complete on a single 8 x H100 80GB GPU instance. Future work should explore the impact of scaling model size and further tune other hyperparameters for multilingual preference optimization.

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A Background on DPO and RLOO

(1) Instruction fine-tuning (SFT) stage: A pre-trained LM is instruction-tuned using a dataset consisting of a given instruction prompt, and (typically) a human-written completion. The LM/policy is trained with a cross-entropy loss over the completion only. Often, the SFT model, denoted as π^{sft} is used to initialize both the reward model (for online RL optimization) and the RLHF policy model.

(2) Preference optimization stage: In this stage, the preference data such as rankings of model responses, are collected through humans or AI feedback. This data is then used to further fine-tune the SFT model (policy) to align the model with human feedback via the collected preferences data. Since collecting human feedback is often very expensive, many preference optimization methods train a separate reward model, on collected preference data to act as a proxy for human preferences, enabling for *online* preference feedback on LLM responses without requiring human intervention. Preference optimization can be performed in a number of ways:

Online Preference Optimization includes training a reward model, typically through binary classification, which is then used to provide online feedback in the optimization of the policy with the following objective:

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [r_{\phi}(x, y) - \beta p_{KL}],$$

with $p_{KL} = D_{KL} \pi_{\theta}(\cdot|x) || \pi_{\text{ref}}(\cdot|x)$

where β is meant to control the distance from the initial policy π_{ref} during the optimization of reward $r_{\theta}(x, y)$ as proposed in (Stiennon et al., 2022). The KL-penalty p_{KL} is crucial as penalty-free optimization of the reward model leads to degradation in the coherence of the model.

Direct Preference Optimization (DPO; Rafailov et al., 2023) collects pairwise preferences (often over LLM responses) and fine-tunes the policy to be implicitly consistent with the collected preference pairs by using the following loss:

$$-\log \sigma(\beta \log \frac{\pi_{\theta}(y_+|x)}{\pi_{\text{ref}}(y_+|x)} - \beta \log \frac{\pi_{\theta}(y_-|x)}{\pi_{\text{ref}}(y_-|x)})$$

Different from the online RL methods, DPO skips the reward modeling and enables preference optimization in an offline manner without requiring

online samples from the policy during the training. At its core, DPO uses the analytical formulation of the canonical KL-controlled RLHF objective detailed in equation A and the assumption that preferences can be modeled by the Bradley-Terry model (Bradley and Terry, 1952), to for-go reward modeling, simplifying the problem into a supervised style classification task.

While reinforcement learning approaches share the components above, techniques differ in formulating the reward and the sample-based loss. In this work, we use **REINFORCE-Leave-one-out (RLOO; Kool et al., 2019; Ahmadian et al., 2024)** estimator as the online preference optimization method as it is effective and more efficient than Proximal Policy Optimization (PPO) (Schulman et al., 2017b). RLOO is a multi-sample extension of REINFORCE (Williams, 1992), where multiple online generations are sampled from the policy per prompt which enables to reduce variance without requiring an additional network as opposed to PPO:

$$\frac{1}{k} \sum_{i=1}^k [R(y_{(i)}, x) - \frac{1}{k-1} \sum_{j \neq i} R(y_{(j)}, x)]$$

$$\nabla \log \pi(y_{(i)}|x) \text{ for } y_{(1)}, \dots, y_{(k)} \stackrel{i.i.d.}{\sim} \pi_{\theta}(\cdot|x)$$

RLOO_k considers each $y_{(i)}$ individually and uses the remaining $k-1$ samples to create an unbiased estimate of the expected return for the prompt, akin to a *parameter-free* value-function, but estimated at each training step.

B Additional Win-Rate Results

B.1 Dolly Open-Ended Instruction Following

Tables 4 and 5 show open-ended generation win-rate scores of preference-trained models on English and unseen languages respectively. Win-rates are measured against the original Aya-23-8B model (Aryabumi et al., 2024).

B.2 RLOO vs DPO

To provide a head-to-head comparison between DPO and RLOO, Table 6 shows the win-rate evaluation between models trained with RLOO method with the models trained with DPO.

B.3 XLSum Summarization

Table 7 shows the win-rate scores of preference-trained models on 15 languages that are covered by our 23 language list. Win-rates are measured against the original Aya-23-8B model (Aryabumi

		English		
		Win%	Loss%	$\Delta W-L\%$
DPO	EN-1	52.0	33.5	18.5
	ML-5	50.5	28.5	22.0
	ML-23	44.5	36.5	8.0
	ML-23*	57.5	31.0	26.5
RLOO	EN-1	47.5	38.5	9.0
	ML-5	55.5	30.5	25.0
	ML-23	53.0	37.0	16.0
	ML-23*	53.0	35.0	18.0

Table 4: English Dolly win-rate results for DPO/RLOO preference optimized Aya 23 8B on multiple training data mixtures: EN-1 (English Only), ML-5 (5 Languages), ML-23 (23 Languages). All runs are done with 50K total training examples with the exception of ML-23*, which is done with 230K total training examples. We report results for the best checkpoint across 2 epochs.

		Avg. Unseen Langs.		
		Win %	Loss %	$\Delta W-L\%$
EN-1	DPO	42.9	40.9	2.0
	RLOO	46.3	39.3	7.3
ML-5	DPO	43.3	39.5	3.8
	RLOO	54.9	35.5	19.4

Table 5: Win-rates for the 22 and 18 languages that are not included in the training data (“unseen”) for EN-1 and ML-5 respectively. We observe cross-lingual transfer from preference optimization, with an increased degree of transfer enhanced by multilingual training.

et al., 2024). The average generation length for the base model, the best DPO, and the best RLOO models are 138, 234, and 171 tokens respectively. Length bias is a known property of DPO (Park et al., 2024) and can bias GPT-4 as an evaluator (Saito et al., 2023) accordingly. Because the base model and the RLOO model generation are similar in length, it is unlikely that the large gains in win-rate for the RLOO model against the base model are caused by GPT-4 as a judge preferring longer responses.

C Discriminative Benchmark Results

D Judge Prompt

System preamble:

You are a helpful following assistant whose goal is to select the preferred (least wrong) output for a given instruction in [LANGUAGE_NAME].

Prompt Template:

Which of the following answers is the best one

for given instruction in [LANGUAGE_NAME]. A good answer should follow these rules:

- 1) It should be in [LANGUAGE_NAME]*
- 2) It should answer the request in the instruction*
- 3) It should be factually and semantically comprehensible*
- 4) It should be grammatically correct and fluent.*

Instruction: [INSTRUCTION]

Answer (A): [COMPLETION A]

Answer (B): [COMPLETION B]

FIRST provide a one-sentence comparison of the two answers, explaining which you prefer and why. SECOND, on a new line, state only ‘Answer (A)’ or ‘Answer (B)’ to indicate your choice. If the both answers are equally good or bad, state ‘TIE’. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <‘Answer (A)’ or ‘Answer (B)’ or ‘TIE’>

E Full Language Set Win-Rates

We provide full win-rate results broken down for all 23 languages for the ML-23-230K DPO run in Table 11 and the the ML-23-230K RLOO run in Table 12

F Language List

We provide a list and description of all languages supported by Aya 23 8B which we use to perform multilingual evaluations in Table 13.

			English			Avg. 23 Langs.		
		Num Examples	Win%	Loss%	$\Delta W-L\%$	Win%	Loss%	$\Delta W-L\%$
RLOO vs DPO	EN-1	50K	37.5	46.5	-9.0	44.2	40.3	4.0
	ML-5	50K	50.5	40.0	10.5	51.4	38.8	12.6
	ML-23	50K	51.0	39.5	11.5	50.4	41.3	9.1
	ML-23	230K	50.0	35.0	15.0	47.4	41.1	6.2

Table 6: Direct win-rate comparisons for RLOO and DPO models on Dolly. RLOO models consistently outperform their DPO counterparts across all dataset splits.

		Average 15 Languages		
		Win%	Loss%	$\Delta W-L\%$
DPO	EN-1	52.5	41.6	10.9
	ML-5	50.5	43.4	7.1
	ML-23	53.0	40.9	12.1
	ML-23*	56.7	37.5	19.2
RLOO	EN-1	58.0	35.8	22.2
	ML-5	65.1	30.2	34.9
	ML-23	70.7	25.3	45.4
	ML-23*	68.9	26.7	42.2

Table 7: 15 language XLSum win-rate results for DPO/RLOO preference optimized Aya 23 8B on multiple training data mixtures: EN-1 (English Only), ML-5 (5 Languages), ML-23 (23 Languages). All runs are done with 50K total training examples with the exception of ML-23*, which is done with 230K total training examples. We report results for the best checkpoint across 2 epochs.

Model	Held out tasks (Accuracy %)			
	XCOPA	XSC	XWG	<u>Avg</u>
Base Aya 23 8B	59.8	62.3	80.7	67.6
DPO Aya 23 8B	59.9	62.6	80.7	67.7
RLOO Aya 23 8B	59.4	62.8	81.1	67.8

Table 8: Results for **discriminative unseen (held-out) task** evaluation. Results are reported as the zero-shot performance averaged across all languages of XCOPA, XStoryCloze, and XWinoGrad. DPO and RLOO checkpoints are for ML-23 230K runs

	en	ar	de	es	fr	hi	id	it	nl	pt	ro	ru	uk	vi	zh	Avg
Base Aya 23 8B	54.6	45.1	50	50.9	51	39.7	48.8	50.7	49.7	50.8	49.9	47.8	46.8	46.5	47.1	48.2
DPO Aya 23 8B	54.9	45.7	50.0	51.1	51.3	40.0	49.0	51.2	49.8	51.1	49.9	48.0	47.0	46.8	47.6	48.5
RLOO Aya 23 8B	54.0	45.2	50.0	50.5	50.4	39.8	48.6	50.3	49.1	50.47	49.48	47.79	46.64	46.49	47.1	48.0

Table 9: **Multilingual MMLU** (5-shot) results for base, DPO, and RLOO Aya 23 models.

	de	en	es	fr	ja	ru	zh	Avg
Base Aya 23 8B	40.4	48.0	45.2	38.8	12.8	38.0	32.8	36.6
DPO Aya 23 8B	39.6	45.6	44.4	41.2	8.4	37.6	35.2	36.1
RLOO Aya 23 8B	39.6	46.4	38.4	39.6	14.0	34.8	33.2	35.1

Table 10: **Multilingual Grade School Math benchmark (MGSM)** results for . We use questions with answers followed by CoT prompt (5-shot) in the same language (native_cot) as the dataset and strict-match score as the evaluation metric.

	Win (%)	Tie (%)	Loss (%)	ΔW-L (%)
en	57.5	11.5	31.0	26.5
vi	54.5	10.5	35.0	19.5
tr	47.5	13.0	39.5	8.0
pt	51.0	7.0	42.0	9.0
de	50.0	10.0	40.0	10.0
ar	50.0	12.0	38.0	12.0
cs	45.0	11.5	43.5	1.5
el	48.0	6.5	45.5	2.5
es	39.5	10.5	50.0	-10.5
fa	51.5	10.5	38.0	13.5
fr	47.0	10.0	43.0	4.0
he	48.5	10.5	41.0	7.5
hi	57.5	10.5	32.0	25.5
id	52.5	13.0	34.5	18.0
it	50.5	11.0	38.5	12.0
ja	53.0	15.5	31.5	21.5
ko	51.0	12.5	36.5	14.5
nl	49.0	14.5	36.5	12.5
pl	50.5	10.0	39.5	11.0
ro	58.0	8.5	33.5	24.5
ru	46.5	8.0	45.5	1.0
uk	52.0	8.0	40.0	12.0
zh	45.0	13.0	42.0	3.0

Table 11: All language results for DPO ML-23-230K

	Win (%)	Tie (%)	Loss (%)	ΔW-L (%)
en	53.0	12.0	35.0	18.0
vi	58.5	6.5	35.0	23.5
tr	54.5	10.0	35.5	19.0
pt	54.5	11.0	34.5	20.0
de	54.0	10.5	35.5	18.5
ar	49.5	12.5	38.0	11.5
cs	57.5	8.0	34.5	23.0
el	50.5	7.0	42.5	8.0
es	55.5	8.0	36.5	19.0
fa	56.0	12.0	32.0	24.0
fr	49.5	8.0	42.5	7.0
he	56.0	8.0	36.0	20.0
hi	62.0	12.0	26.0	36.0
id	49.5	9.5	41.0	8.5
it	51.0	10.0	39.0	12.0
ja	58.5	10.5	31.0	27.5
ko	50.5	9.5	40.0	10.5
nl	49.0	10.0	41.0	8.0
pl	52.5	5.5	42.0	10.5
ro	54.0	11.0	35.0	19.0
ru	51.5	6.0	42.5	9.0
uk	50.0	14.0	36.0	14.0
zh	50.5	10.0	39.5	11.0

Table 12: All language results for RLOO ML-23 230K

Code	Language	Script	Family	Subgrouping
ar	Arabic	Arabic	Afro-Asiatic	Semitic
cs	Czech	Latin	Indo-European	Balto-Slavic
de	German	Latin	Indo-European	Germanic
el	Greek	Greek	Indo-European	Graeco-Phrygian
en	English	Latin	Indo-European	Germanic
es	Spanish	Latin	Indo-European	Italic
fa	Persian	Arabic	Indo-European	Iranian
fr	French	Latin	Indo-European	Italic
he	Hebrew	Hebrew	Afro-Asiatic	Semitic
hi	Hindi	Devanagari	Indo-European	Indo-Aryan
id	Indonesian	Latin	Austronesian	Malayo-Polynesian
it	Italian	Latin	Indo-European	Italic
jp	Japanese	Japanese	Japonic	Japanesic
ko	Korean	Hangul	Koreanic	Korean
nl	Dutch	Latin	Indo-European	Germanic
pl	Polish	Latin	Indo-European	Balto-Slavic
pt	Portuguese	Latin	Indo-European	Italic
ro	Romanian	Latin	Indo-European	Italic
ru	Russian	Cyrillic	Indo-European	Balto-Slavic
tr	Turkish	Latin	Turkic	Common Turkic
uk	Ukrainian	Cyrillic	Indo-European	Balto-Slavic
vi	Vietnamese	Latin	Austroasiatic	Vietic
zh	Chinese	Han & Hant	Sino-Tibetan	Sinitic

Table 13: 23 languages supported in Aya 23 model (Aryabumi et al., 2024) with each language’s script, family, and subgrouping